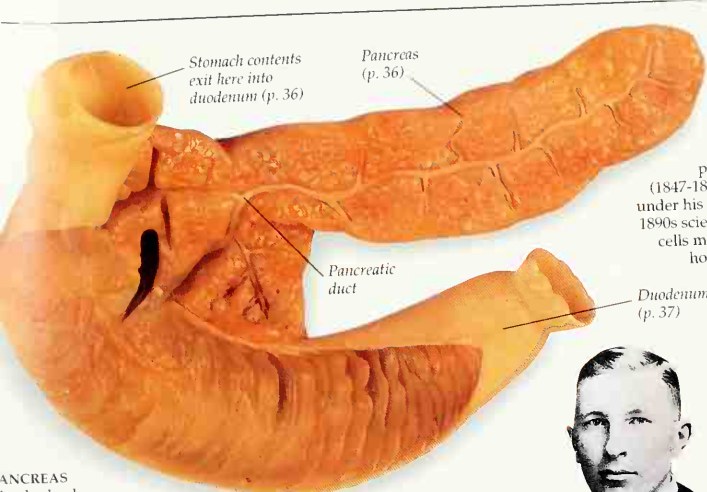
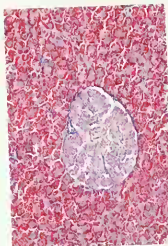




PANCREAS
The dual-role pancreas is both a ducted or "exocrine" gland and a ductless or "endocrine" gland. It produces powerful juices for digestion (p. 36), which pass along the pancreatic duct into the intestine. Its other role is making two hormones, insulin and glucagon. Hormones do not ooze along ducts. They seep directly into blood (pp. 32-33) flowing through the organ, past the manufacturing cells. The balance of insulin and glucagon controls the level of glucose (sugar) in the blood.

PANCREATIC ISLETS
The pancreas (p. 36) is dotted with more than 1 million tiny clumps of cells known as islets of Langerhans; one is shown, right. They are named after the German physician Paul Langerhans (1847-1888), who discovered them under his microscope in 1869. In the 1890s scientists discovered that islet cells made secretions, later called hormones, including insulin.



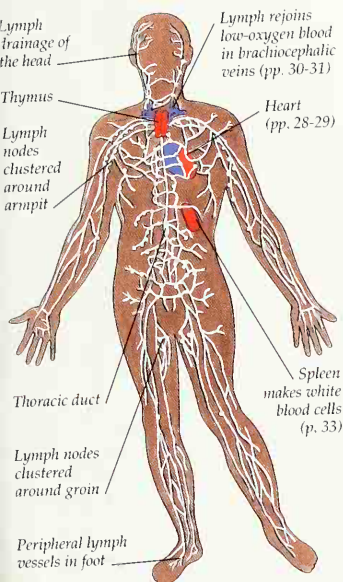
Sir Frederick Banting
(1891-1941)



Charles Best
(1899-1978)

THE INSULIN STORY

The hormone insulin tells the liver to change glucose into starch and store it. In 1922 Canadian Frederick Banting and American Charles Best announced that they had extracted insulin and tested it on dogs with diabetes, a serious condition caused by lack of insulin. Banting and laboratory director J. Macleod received a Nobel Prize in 1923.



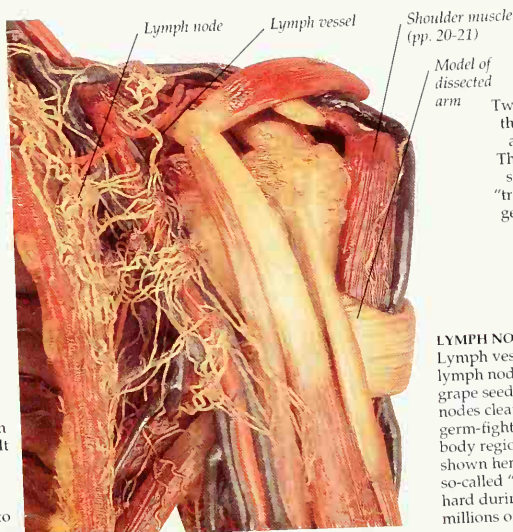
The lymphatic system

Lymph is a milky liquid that begins as watery fluids that have collected in spaces around cells (pp. 12-13), and watery fluids that have leaked from small blood vessels (pp. 30-31). These fluids channel slowly into a body-wide system of veinlike tubes, with one-way valves. Lymph has several jobs: it collects wastes from cell activity; it distributes nutrients, especially fats; it also carries white blood cells (p. 33) around the body in the fight against infection.



THYMUS GLAND

Two main glands are associated with the lymph system: the spleen (p. 32) and the thymus in the upper chest. The thymus is large in childhood but shrinks during adult life. It helps to "train" white blood cells to recognize germs and other body invaders. The electron microscope shows the cortex (outer area) of the thymus. The spheres are a type of white blood cell, called lymphocytes.



LYMPH NODES

Lymph vessels widen at intervals to form lymph nodes, which vary from the size of grape seeds to larger than whole grapes. The nodes clean and filter the lymph and store germ-fighting white blood cells. In several body regions, nodes are clustered together, as shown here in the armpit. These nodes are the so-called "glands" that swell up and become hard during illness, because they contain millions of white blood cells and dead germs.

SLOW CIRCULATOR

The system of lymph vessels is almost as extensive as the blood system, but it is only "half" a circulation as it has no pump. A person has about 1.8 pt (1 liter) of lymph, on average. It oozes along its vessels, propelled indirectly by the pressure of the blood, and by the body's muscles which massage the vessel walls. The major lymph vessels empty their fluid back into the blood in the main veins near the heart.